Technology Readiness Index for M&E & Regional Benchmarking Framework.

Cross-country impact measurement and intervention-sequencing framework

Author: Swaraj Singh | Organization of Southern Cooperation

Note on redactions and masking

This document is a **redacted version** of an analytical framework developed during my tenure at the Organization of Southern Cooperation (OSC). It is shared as part of my job application to evidence experience with impact measurement framework development, cluster-based benchmarking, and decision-support modelling.

layers of confidentiality protection are in place:

- **(a) Indicator masking.** The eleven composite indicators used to construct the three thematic pillars are referred to throughout this document by anonymised codes `T_1, T_2, T_3, T_4` for the Technology Infrastructure & Access pillar; `I_1, I_2, I_3, I_4` for the Innovation Ecosystem pillar; and `P_1, P_2, P_3` for the Policy & Regulatory Environment pillar. The specific identity and source attribution of each indicator are not disclosed.

What is preserved: the full methodological pipeline (Z-score standardisation, MICE iterative imputation, Gaussian Mixture Model clustering, ANOVA validation, PCA weight derivation, composite scoring formula, Priority Matrix logic); the analytical visualisations from the framework, generated on the masked data; the pillar names themselves, which reflect standard categories in the digital-readiness literature and are not proprietary; and one fully worked country snapshot (Afghanistan) demonstrating the structure and depth of the framework's per-country outputs.

1. Introduction

In an era marked by rapid technological advancement and increasing global interconnectivity, the imperative for nations particularly those in the Global South to adapt, innovate, and thrive has never been more pressing. The Fourth Industrial Revolution offers unprecedented opportunities, but also exacerbates existing inequalities. A tailored, context-specific approach to technological development in these nations is not only an economic imperative but a moral one, aligning with global commitments to sustainable development and equitable progress.

This framework was developed to provide a structured, data-driven understanding of the technological landscape across member states of the Organization of Southern Cooperation. It is designed to function simultaneously as an impact measurement tool establishing a comparable baseline against which progress can be tracked and as a decision-support model for the sequencing and prioritisation of technical assistance interventions across heterogeneous country contexts.

Objectives

- **Mapping the technological landscape:** Provide a comprehensive, multidimensional assessment of digital infrastructure, innovation ecosystems, and policy environments across the country set.
- **Country-specific nuance:** Move beyond aggregate scores to expose intra-cluster heterogeneity and the specific gaps and opportunities each country presents.
- **Strategic framework for development:** Establish a flexible, replicable framework that can underpin future strategies for endogenous technological growth, digital literacy, and quality-of-life improvement.

2. Analytical Framework Overview

The framework is structured around three thematic pillars, selected to capture the principal dimensions along which a country's technological readiness can be evaluated. Within each pillar, a curated set of composite indicators from established multilateral sources is used to mitigate primary data scarcity each underlying composite index already subsumes multiple sub-indicators, providing aggregated insight that would otherwise require primary data collection at infeasible scale.

Pillar	Captures	Indicator codes
Technology Infrastructure & Access	Foundational digital connectivity, government digital service delivery, cybersecurity posture, and population-level digital penetration.	T_1, T_2, T_3, T_4
Innovation Ecosystem	National productive capacity, R&D intensity, human capital development, and public investment in education.	I_1, I_2, I_3, I_4
Policy & Regulatory Environment	Business operating environment, presence of emerging-technology regulation, and government-technology enabling conditions.	P_1, P_2, P_3

Total: **11 composite indicators** across three pillars. Specific indicator identities and source attributions are redacted; codes are used throughout the document as introduced in the note above.

3. Methodology

The methodology is structured as a sequential analytical pipeline. Each stage produces an output that feeds the next, and each is independently validated before progressing.

3.1 Data preprocessing

3.1.1 Standardisation

Indicators in the dataset are on heterogeneous scales: some are rankings (ordinal, inverse), some are percentages, others are index scores on bounded continuous scales. To enable aggregation and meaningful comparison, Z-score standardisation is applied to every variable:

$$Z = (X - \mu) / \sigma$$

where X is the original observation, μ is the variable mean across the country set, and σ is the standard deviation. Post-standardisation, each value represents the number of standard deviations the country lies from the cross-country mean for that indicator. The procedure preserves rank order and inter-country relationships within each variable while placing all variables on a common scale, allowing valid aggregation downstream.

3.1.2 Missing data treatment

Data availability is a persistent challenge for the country set in scope. Listwise deletion would eliminate a substantial fraction of country-indicator observations and bias downstream estimates; mean imputation would compress variance artificially. Instead, missing values are handled via Multivariate Imputation by Chained Equations (MICE) iterative imputation, which uses the observed relationships between variables to predict missing values across multiple iterations, producing imputed values that preserve the multivariate covariance structure of the observed data.

3.2 Cluster Analysis Gaussian Mixture Model

To impose structure on the cross-country variation before scoring, a Gaussian Mixture Model (GMM) clustering analysis is performed on the standardised indicator matrix. GMM is a probabilistic clustering method that models the data as a mixture of a finite number of Gaussian distributions; it produces a soft assignment, returning for each country the posterior probability of membership in each cluster. GMM is chosen over hard-partition methods (k-means) because it accommodates non-spherical cluster shapes which better matches the actual covariance structure observed in the standardised data and because the probabilistic assignment exposes borderline cases, which are analytically important for countries near the boundary between development trajectories.

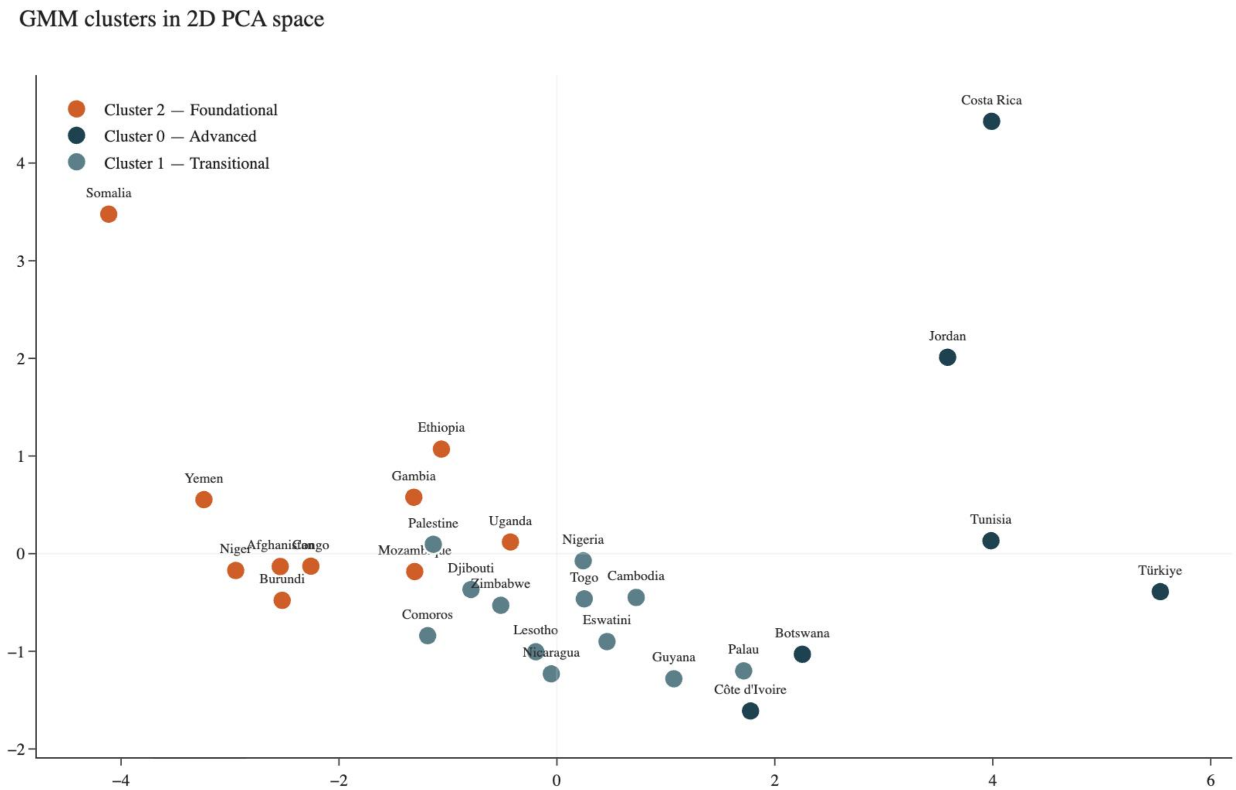
A three-component GMM identifies three distinct clusters, interpretable as tiers of technological readiness:

- **Cluster 0 Advanced:** Above-average performance across most indicators, with mature digital infrastructure, established innovation policy, and stable regulatory environments.
- **Cluster 1 Transitional:** Moderate average performance with substantial intra-cluster variance; countries in this cluster face heterogeneous combinations of strengths and gaps and require disaggregated rather than uniform intervention design.
- **Cluster 2 Foundational:** Lower performance across most indicators, characterised by limited digital infrastructure, low R&D intensity, and underdeveloped emerging-technology regulation.

Visual validation in 2D PCA space

Principal Component Analysis (PCA) is used to project the eleven-dimensional standardised indicator space onto two axes capturing the maximum variance. Plotting the GMM cluster assignments in this projection provides a visual robustness check if the clusters genuinely separate, they should separate visibly here.

[Figure: 2D PCA cluster scatter]



PC1 and PC2 together capture approximately 65% of total variance across the indicator set. The three clusters separate cleanly along PC1, which is interpretable as a composite axis of overall technological readiness, with Cluster 0 countries occupying the upper-right region and Cluster 2 countries the lower-left.

Geographic distribution of clusters

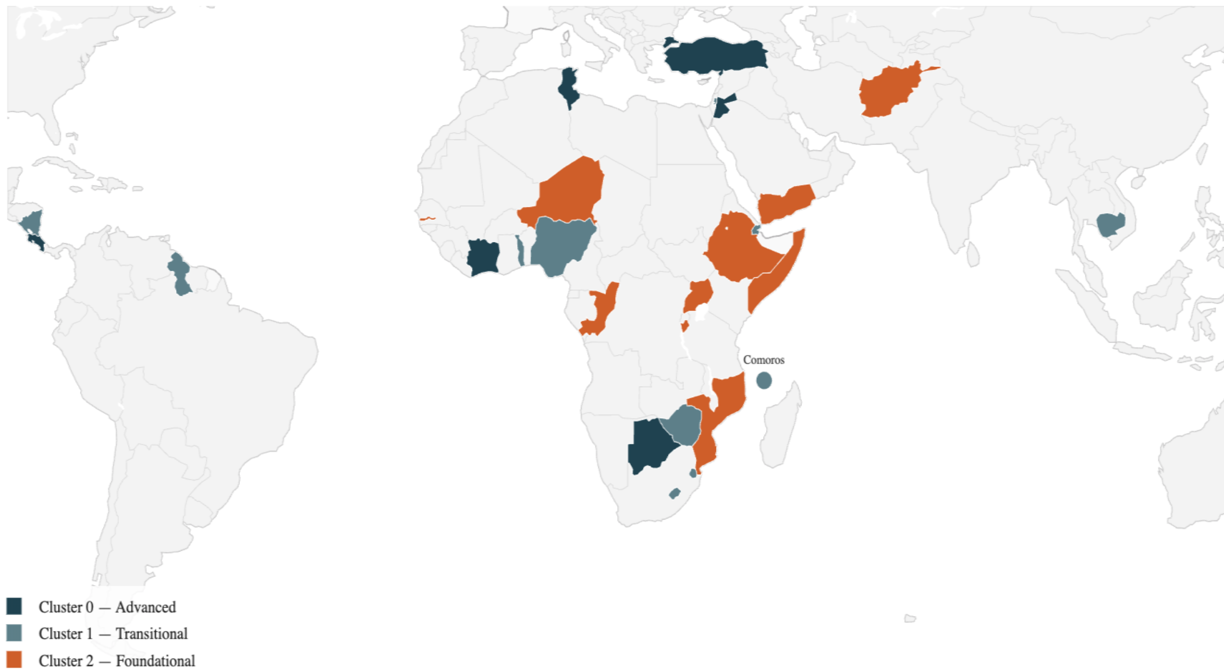


Figure 2 Geographic distribution of cluster assignments across the country set.

The geographic projection surfaces the regional dimensions of the cluster structure. The Advanced tier (Cluster 0) is broadly distributed including countries from Latin America (Costa Rica), West Africa (Côte d'Ivoire), Southern Africa (Botswana), the Middle East / North Africa (Jordan, Tunisia), and the broader Mediterranean basin (Türkiye). The Foundational tier (Cluster 2) concentrates heavily in the Horn of Africa and the Sahel, with extensions to Central Africa and parts of South Asia. The Transitional tier (Cluster 1) is the most geographically dispersed, spanning small island developing states in the Pacific (Palau), the Caribbean basin (Guyana, Nicaragua), Southeast Asia (Cambodia), and Southern Africa (Eswatini, Lesotho, Zimbabwe). Tiny states are rendered with proxy markers where their geometries are too small to display at continental scale.

3.3 Correlation Analysis

A pairwise correlation matrix is computed across the eleven PCA-weighted indicator scores. The exercise serves two purposes: to surface multicollinearity that could distort downstream weighting, and to verify that indicators within the same pillar behave as theoretically expected that is, that intra-pillar correlations are positive and that cross-pillar correlations reflect plausible substantive linkages.

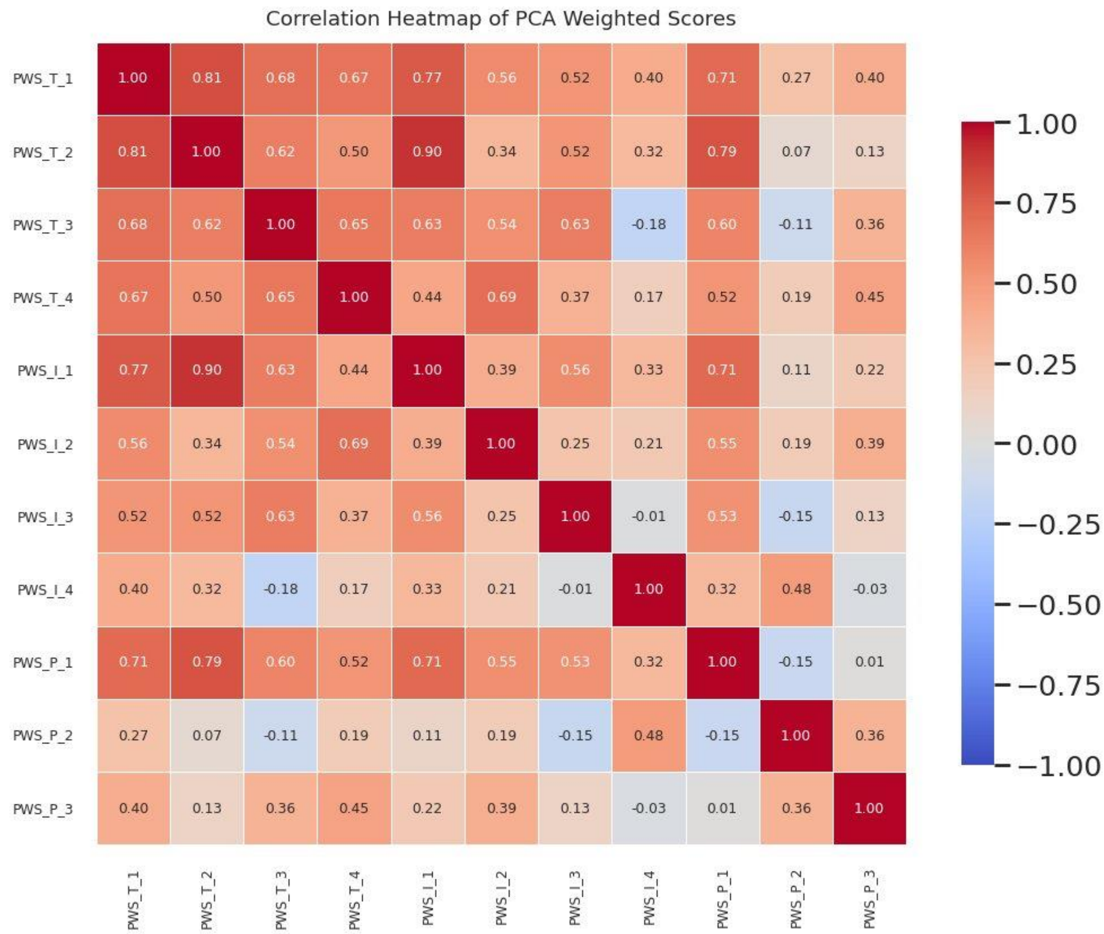


Figure 3 Pairwise correlation matrix of PCA-weighted indicator scores.

The observed correlation structure is consistent with theoretical expectation. The strongest correlation in the matrix is between `T_2` and `I_1` (r = 0.90), reflecting the deep complementarity between digital penetration and broader productive capacity. `T_1` correlates strongly with `T_2` (r = 0.81), indicating that the leading technology-pillar indicators move together as a coherent bundle. Cross-pillar correlations are mostly positive but moderate, confirming that the three pillars are related but not redundant a desirable property for a multidimensional composite framework. `I_4` and `P_2` show the weakest connections to the rest of the indicator set, foreshadowing their later identification as non-discriminating in the ANOVA validation.

3.4 Cluster robustness ANOVA validation

If clusters arbitrarily carve up a continuous distribution, they do not capture meaningful structure. A one-way ANOVA is conducted for each indicator with cluster membership as the categorical predictor; the null hypothesis is that the population mean of the indicator is equal across the three clusters. Rejection of the null indicates that the clusters capture genuine indicator-level differentiation.

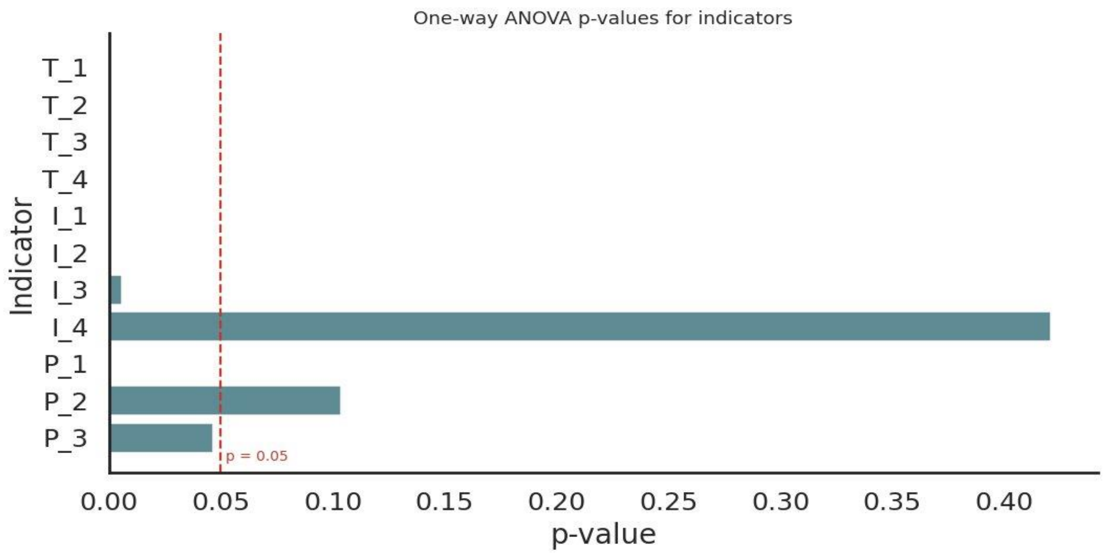


Figure 4 ANOVA p-values per indicator. Red dashed line marks p = 0.05 threshold.

Null Hypothesis (H0): There is no significant difference in the means of the variable being tested (e.g., "Scores_Frontier_tech_readiness_index," "E_Gov_Dev_Index_Rankings," etc.) among the different clusters or groups of countries.

The results of the ANOVA test confirmed that there were significant differences between the clusters across most of the variables, as evidenced by the low P-values (below 0.05) and high F-statistics. This validation reinforced the robustness of the clustering and justified the selection of variables used in the analysis. It also highlighted the significant disparities in technological readiness and development across the target countries, underscoring the need for tailored policy interventions to address the unique challenges and opportunities faced by each cluster of countries.

Furthermore, the results underscore the importance of using composite indices in the analysis. As these indices encapsulate a range of individual indicators, they provide a comprehensive view of the technological landscape in each country, especially when data on individual indicators might be scarce or unavailable.

Nine of the eleven indicators reject the null at $p < 0.05$, with seven rejecting at $p < 0.001$. The two non-significant indicators `I_4` ($p = 0.42$) and `P_2` ($p = 0.10$) are retained in the scoring pipeline on substantive grounds, as both capture analytically important dimensions even where they do not differentiate strongly between clusters. The full ANOVA results are tabulated below.

Indicator	F-statistic	p-value	Significance
T_1	26.180	<0.001	***
T_2	30.628	<0.001	***
T_3	23.880	<0.001	***
T_4	9.919	0.0007	***
I_1	28.490	<0.001	***
I_2	24.256	<0.001	***
I_3	6.349	0.0059	**
I_4	0.896	0.4207	n.s.
P_1	12.925	0.0001	***
P_2	2.483	0.1038	n.s.
P_3	3.477	0.0465	*

*Table 1 One-way ANOVA results per indicator (significance codes: *** p<0.001, ** p<0.01, * p<0.05, n.s. not significant).*

3.5 PCA Weight Assignment

Following the ANOVA analysis, a Principal Component Analysis (PCA) was conducted to assign weights to the selected indicators based on their contribution to the variance within the dataset. The normalized PCA weights were computed, which reflect the relative importance of each indicator in explaining the variance observed in the technological readiness across the target countries.

Equal weighting across indicators treats every variable as equally informative a position not supported by the data. Instead, weights are derived from the loadings of the first principal component (PC1), which by construction captures the dominant direction of variance in the standardised indicator space. Indicators with stronger PC1 loadings carry proportionally more weight; the resulting weights are normalised to sum to one.

PC1 captures 49.0% of total variance in the indicator set a strong dominant dimension consistent with the cluster-tier interpretation visible in the 2D projection. The derived weights are presented below.

Indicator	PCA Weight
T_1	0.1278
T_2	0.1196
T_3	0.1117
T_4	0.1059
I_1	0.1186
I_2	0.0942
I_3	0.0896
I_4	0.0433
P_1	0.1147
P_2	0.0207
P_3	0.0538

Table 2 Normalised PCA-derived weights per indicator. Weights sum to 1.000.

Two analytically important observations: First, the two indicators flagged as non-discriminating in the ANOVA validation (`I_4` and `P_2`) are also the two with the lowest PCA weights confirming consistency between the two independent diagnostics. Second, the dominant weights cluster in the Technology Infrastructure & Access and Innovation Ecosystem pillars, reflecting the structure of variance in the underlying data: these are the dimensions along which countries differ most.

Here are the key significances of employing PCA weights in our analysis:

1. Dimensionality Reduction:

PCA aids in reducing the dimensionality of the data while retaining the most critical information. This is particularly beneficial in our case, given the diverse set of indicators, as it allows for a more manageable, yet insightful, analysis.

2. Variable Importance:

The PCA weights provide a quantitative measure of the relative importance of each indicator. This is instrumental in understanding which aspects of technological readiness are more pronounced and potentially more impactful.

3. Data Interpretability:

By assigning weights to the indicators, PCA enhances the interpretability of the data. It provides a clear hierarchy of indicators based on their contribution to the variance, which is crucial for a more structured and informed analysis.

4. Noise Reduction:

PCA helps in filtering out noise by focusing on the principal components that capture the most variance. This leads to a more robust analysis, minimizing the impact of potential outliers or less relevant variables.

5. Enhanced Visualization:

The reduced-dimension data, along with the PCA weights, facilitates better visualization and understanding of the clusters and patterns within the data, which is crucial for effectively communicating the findings to stakeholders.

3.6 Composite Score Calculation

Pillar composite scores are computed by aggregating the weighted standardised scores of the constituent indicators:

Pillar Composite = Σ (Standardised Indicator Score × PCA Weight)

Three pillar composites are produced per country (Technology Infrastructure, Innovation Ecosystem, Regulatory Environment); a fourth overall composite is computed as the equally-weighted mean of the three pillar composites. Because PCA weights sum to 1.0 across the full eleven-indicator set rather than within each pillar, the pillars produce composites with different natural scales Technology Infrastructure tends to range most widely, while Regulatory Environment composites are more compressed. This asymmetry is methodologically deliberate: it reflects the actual variance distribution in the data and is preserved in the Priority Matrix logic, where comparisons happen within-pillar against cluster averages rather than across pillars.

4. Cross-Country Comparative Analysis

4.1 Country index across pillars and overall composite

Country rankings on each pillar and on the overall composite surface the structure of relative performance across the country set. A country's rank position on a given pillar (1 = highest, 28 = lowest) provides an immediate, scale-free comparator that does not require interpreting raw Z-score magnitudes.

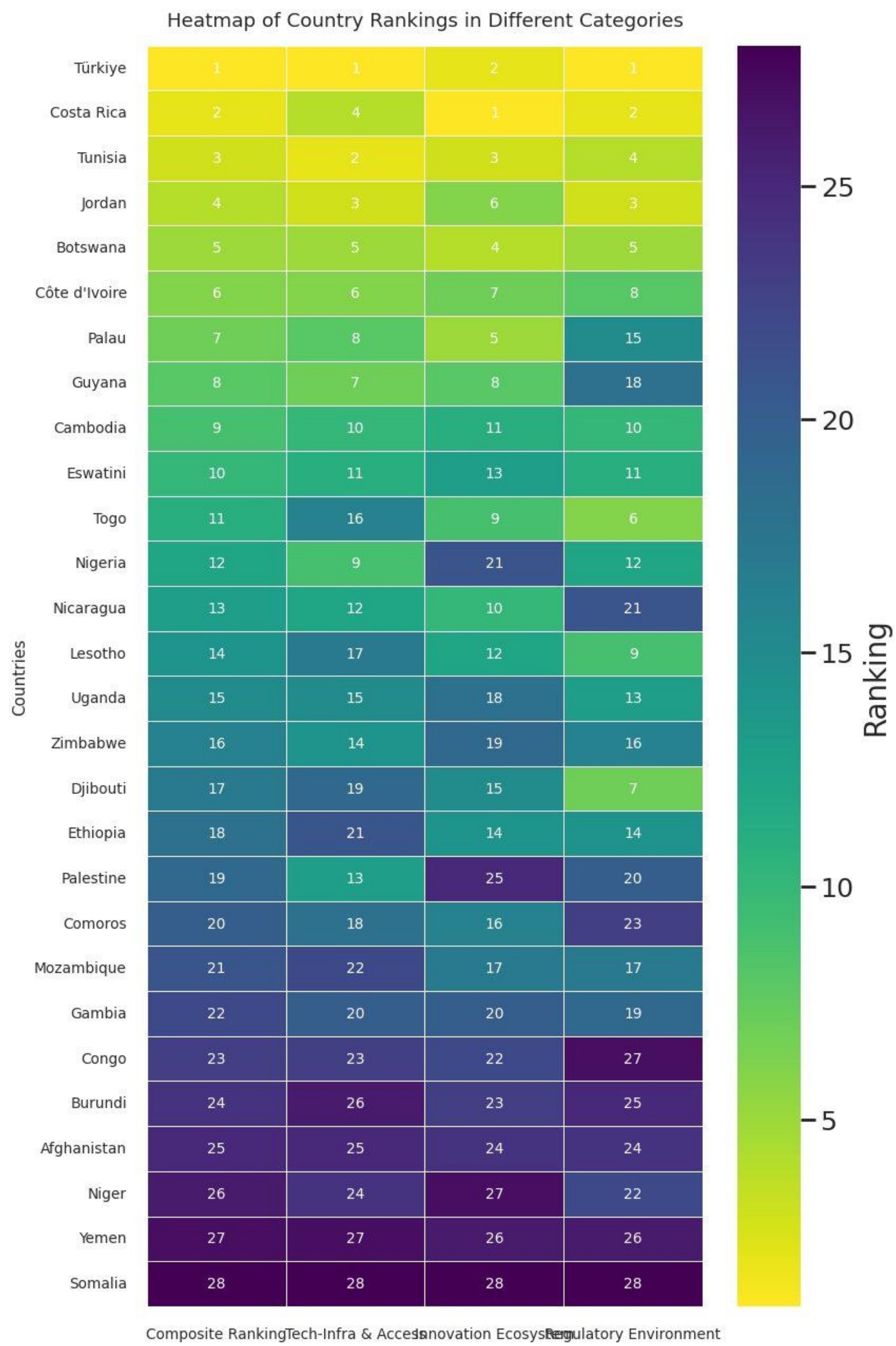


Figure 5 Country rankings across pillars and overall composite. Rows sorted by Composite Ranking (best at top).

The ranking heatmap exposes the country set's tiered structure at a glance. The top stratum Türkiye, Costa Rica, Tunisia, Jordan, Botswana, Côte d'Ivoire leads consistently across all four ranking dimensions. The bottom stratum Niger, Yemen, Somalia trails consistently. Between these poles, ranking divergence is informative: a country whose overall ranking is anchored by strong performance on one pillar but weak performance on another (e.g., Palau, ranked 7th overall but 15th on Regulatory Environment) presents a different intervention profile than one with uniformly middling scores. Such within-country dispersion between pillar ranks is the analytically relevant signal for portfolio sequencing.

4.2 PCA-weighted scores across all countries

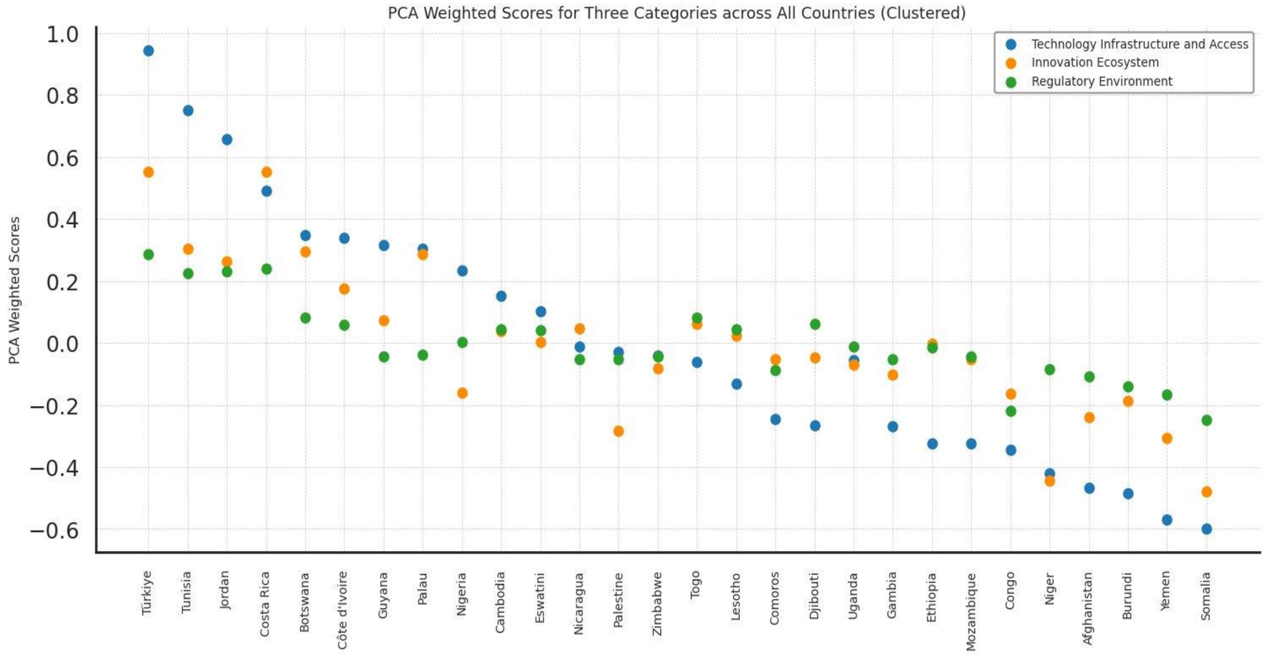


Figure 6 PCA-weighted scores per country across the three pillars. Countries ordered by Technology Infrastructure & Access score descending.

This scatter view elucidates a multi-dimensional perspective on technological readiness across the country set, segmented into the three pillars, encapsulating substantial information in a single visual representation. The plot facilitates several analytical operations simultaneously:

- **Cluster-wise disaggregation.** The PCA-weighted scores can be read in cluster bands, revealing the relative standing of each cluster across the three pillars and providing a nuanced understanding of cross-tier differentiation in technological readiness.
- **Inter-cluster variability.** The plot makes visible the variability among clusters within each pillar, which is essential for identifying the relative strengths and weaknesses of each tier in the technological domain particularly the substantial range within the Transitional tier on the Innovation Ecosystem and Policy & Regulatory dimensions.
- **Indicator resonance across pillars.** By comparing the three coloured series, the chart reveals the resonance or dissonance of pillar performance within each country. Türkiye and Costa Rica show pillar-uniformity; Palau and Côte d'Ivoire show pillar-divergence, with Tech Infrastructure outpacing other pillars substantially.
- **Comparative analysis.** The chart facilitates direct comparative analysis of PCA-weighted scores across all countries and clusters, which is crucial for benchmarking and understanding the competitive landscape of technological readiness in the Global South context.
- **Benchmarking opportunity.** The chart provides a benchmarking opportunity for individual countries to compare their pillar profiles against others within the same cluster or across different clusters important for setting realistic, peer-anchored targets and formulating strategies to bridge identified gaps.

4.3 Parallel coordinates view

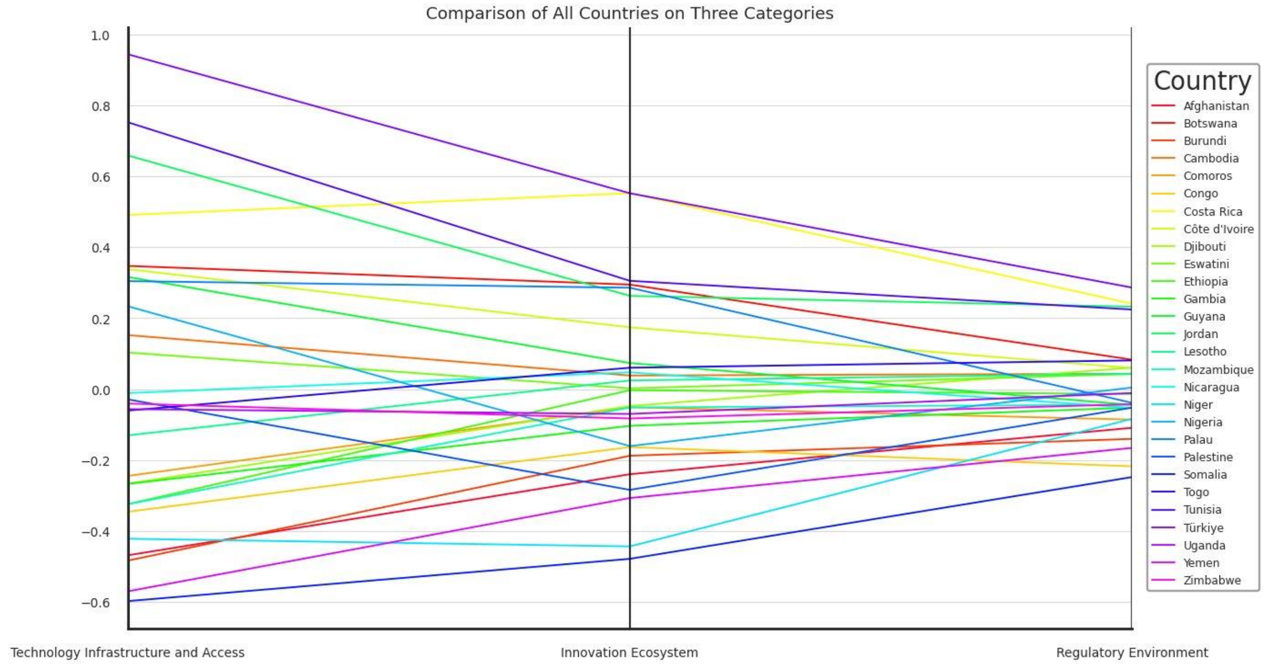


Figure 7 Parallel coordinates plot. Each line is one country, traced across the three pillars.

The parallel coordinates view reveals one of the most analytically consequential patterns in the framework. Reading the chart horizontally, an unmistakable convergence-divergence pattern emerges across the three pillars: countries spread most widely on Technology Infrastructure & Access, narrow somewhat on Innovation Ecosystem, and converge tightly on Regulatory Environment.

The parallel coordinates plot provides a multidimensional comparative visualization of the technological readiness landscape across the target countries. Unlike isolated rankings or unidimensional indicators, this visualization captures the structural balance or imbalance within national technological ecosystems. Each colored line represents an individual country, tracing its relative performance across the three foundational dimensions underpinning endogenous technological development:

- Technology Infrastructure & Access,
- Innovation Ecosystem,
- Regulatory Environment.

Key Analytical Insights

The visualization reveals substantial heterogeneity across the target countries, particularly within the dimension of technological infrastructure and access, where the spread between countries is most pronounced. The wide dispersion observed along this axis indicates significant disparities in foundational technological capabilities, including digital access, connectivity, and broader infrastructure readiness. Top-cluster countries such as Türkiye, Tunisia, Jordan, and Costa Rica exhibit relatively elevated and stable trajectories across all three dimensions.

In contrast, lower-readiness countries display either consistently negative scores or volatile trajectories between dimensions, indicating fragmented ecosystems, limited absorptive capacity, and uneven policy environments where isolated progress in one domain has not yet translated into systemic transformation. The broader policy implication is that countries may require fundamentally different intervention sequencing infrastructure-led, governance-led, or balanced ecosystem-building depending on the specific shape of their structural imbalance, rather than a one-size-fits-all approach.

The Policy and Regulatory frameworks is a kind of point of convergance among the member countries, it suggests that the member countries are showing a greater degree of similarity or convergence at the institutional and policy level than at the infrastructural level. Their regulatory and governance frameworks are comparatively less differentiated.

5. Country Snapshot Afghanistan

This section presents a fully worked country snapshot for **Afghanistan**, chosen as a representative case from the Foundational tier (Cluster 2). The snapshot follows the standard structure applied to each country in the framework: indicator scorecard, performance descriptors by pillar, Priority Matrix, and three comparative visualisations situating the country relative to its cluster peers and across clusters. Gap calculations and the score-vs-cluster-average table are omitted from this redacted version, as they would reveal the scoring methodology directly; the Priority Matrix is presented in visual form only.

5.1 Indicator Scorecard

Standardised Z-scores for Afghanistan across the eleven masked indicators. Cluster assignment: **Cluster 2 Foundational**.

Indicator	Standardised Z-score
T_1	-0.9499
T_2	-0.8172
T_3	-1.1043
T_4	-1.1909
I_1	-0.8128
I_2	-0.7517
I_3	-0.7505
I_4	-0.1271
P_1	-0.6613
P_2	-0.4741
P_3	-0.4331

Table 3 Indicator scorecard for Afghanistan (masked codes; standardised Z-scores).

5.2 Performance on Key Categories

Qualitative descriptors per indicator, derived from the Z-score bins defined in the methodology. The qualitative bands are: Very High (Z ≥ 1.5), High (0.75 to 1.5), Above Average (0.25 to 0.75), Moderate (−0.25 to 0.25), Low (−0.75 to −0.25), Very Low (−1.5 to −0.75), Extremely Low (Z < −1.5).

Pillar	Indicator	Performance Range
Technology Infrastructure & Access	T_1	Very Low
	T_2	Very Low
	T_3	Very Low

	T_4	Very Low
Innovation Ecosystem	I_1	Very Low
	I_2	Very Low
	I_3	Very Low
	I_4	Moderate
Policy & Regulatory Environment	P_1	Low
	P_2	Low
	P_3	Low

Table 4 Qualitative performance descriptors per indicator for Afghanistan.

5.3 Priority Matrix

The Priority Matrix translates the within-country gap analysis into a sequencing recommendation. For each country, the gap between its pillar score and its cluster's average pillar score is computed across all three pillars; the pillar with the largest negative gap receives an Immediate priority, the second-largest receives Medium, and the smallest receives Low. The matrix is presented below in visual form only; the underlying score-vs-cluster-average gap table is omitted in this redacted version.

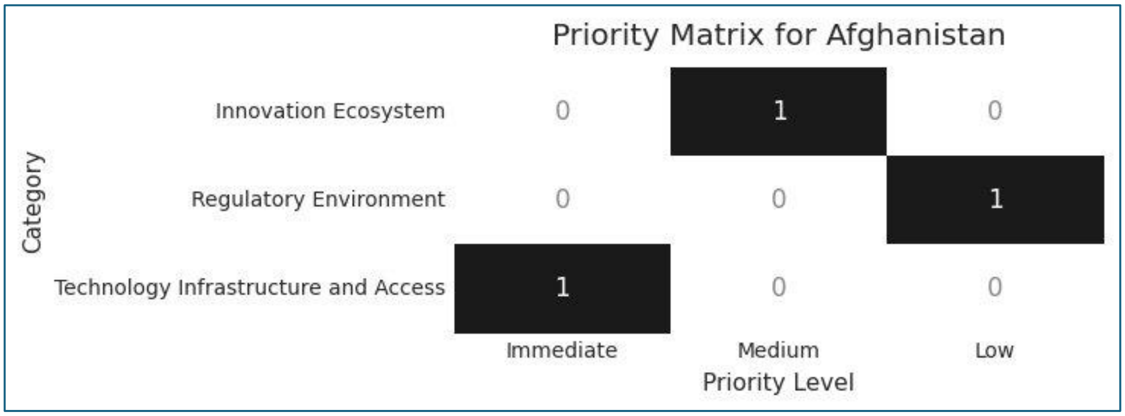


Figure 8 Priority Matrix for Afghanistan.

The matrix identifies Technology Infrastructure & Access as the Immediate priority for Afghanistan, Innovation Ecosystem as the Medium-priority area, and Policy & Regulatory Environment as the Low-priority area. The interpretation is cluster-relative: even though Afghanistan trails on all three pillars in absolute terms, the largest gap to its cluster peers is on the Technology Infrastructure dimension, indicating where peer-anchored short-to-medium-term intervention is both most consequential and most feasible.

5.4 Inter-Cluster Comparative Analysis

This view situates Afghanistan against the average pillar profiles of all three clusters simultaneously, providing both peer-relative and cross-tier reference points.

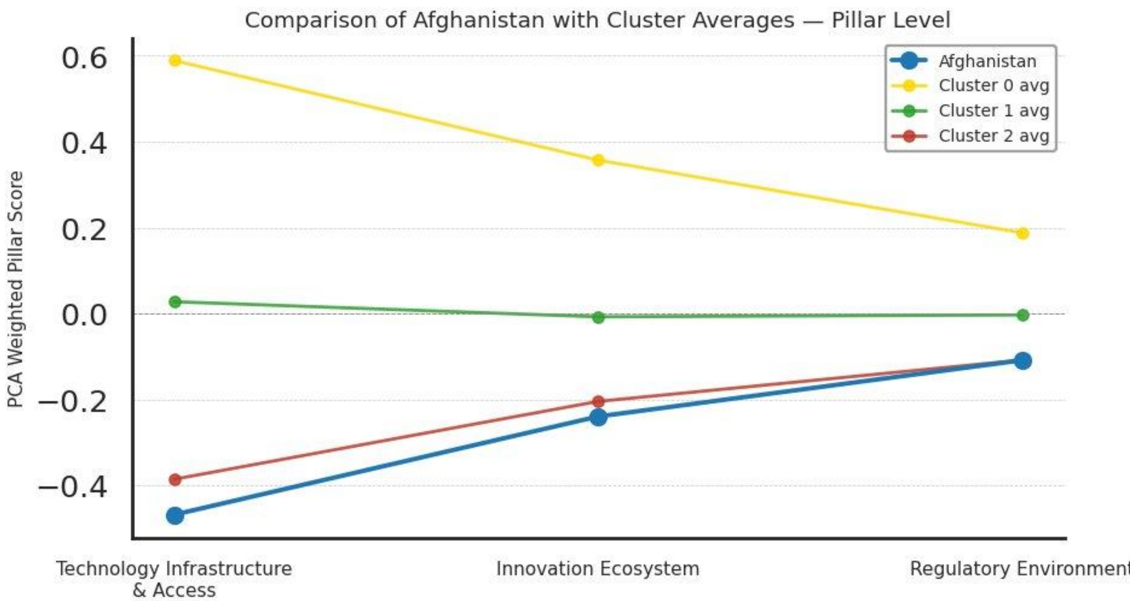


Figure 9 Afghanistan compared to all three cluster averages on the three pillar composites.

The visualisation makes visible the distance between Afghanistan's pillar profile and each of the three cluster averages. Afghanistan trails its own cluster average (Cluster 2) on Technology Infrastructure most clearly, and trails the Cluster 0 (Advanced) average by a substantial margin across all three pillars. The shape of the country line declining from Technology Infrastructure through Innovation to Regulatory Environment mirrors the shape of the Cluster 0 average line, suggesting that the country's relative position-within-profile is similar to advanced-tier countries even where the absolute level is lower.

5.5 Intra-Cluster Comparative Position

This view situates Afghanistan within its own cluster (Cluster 2), showing the distribution of indicator scores across all cluster members. The red marker indicates Afghanistan's position; green markers represent other cluster members.

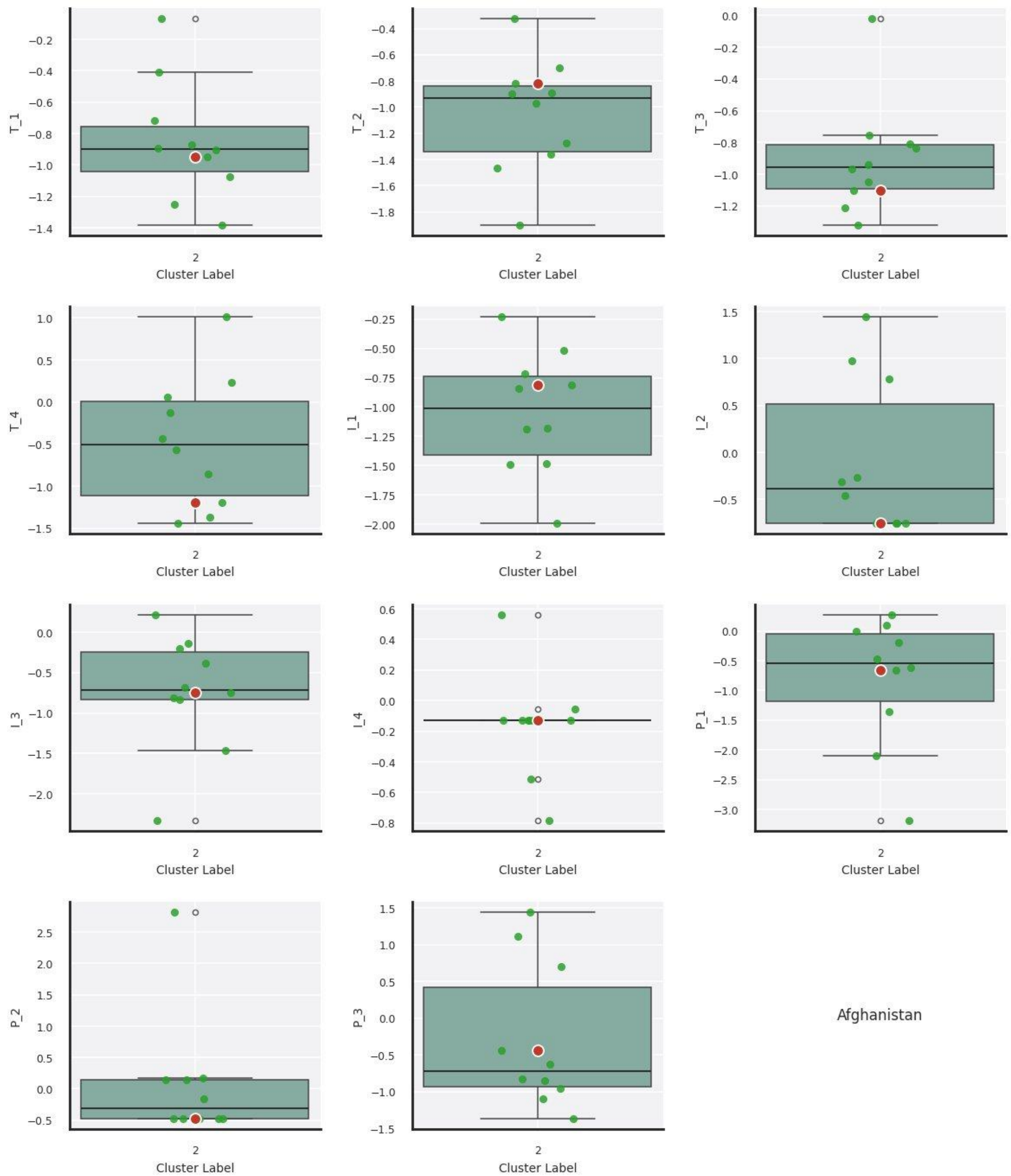


Figure 10 Intra-cluster position of Afghanistan across the eleven indicators (Cluster 2 peers shown in green; Afghanistan in red).

Reading the panels, Afghanistan sits at or below the median of its cluster peers on most indicators, with particularly weak positioning on T_4 (lowest in the cluster) and competitive positioning on I_4 (close to the cluster median). On P_2, Afghanistan falls into the lower mode of the cluster's bimodal distribution.

These intra-cluster patterns are operationally important they identify which dimensions of Afghanistan's underperformance are structural to the foundational tier and which are country-specific gaps relative to similarly-positioned peers.

5.6 Indicator Profile Radar View

The radar view provides a single-glance comparison of Afghanistan's full eleven-indicator profile against the average profiles of all three clusters.

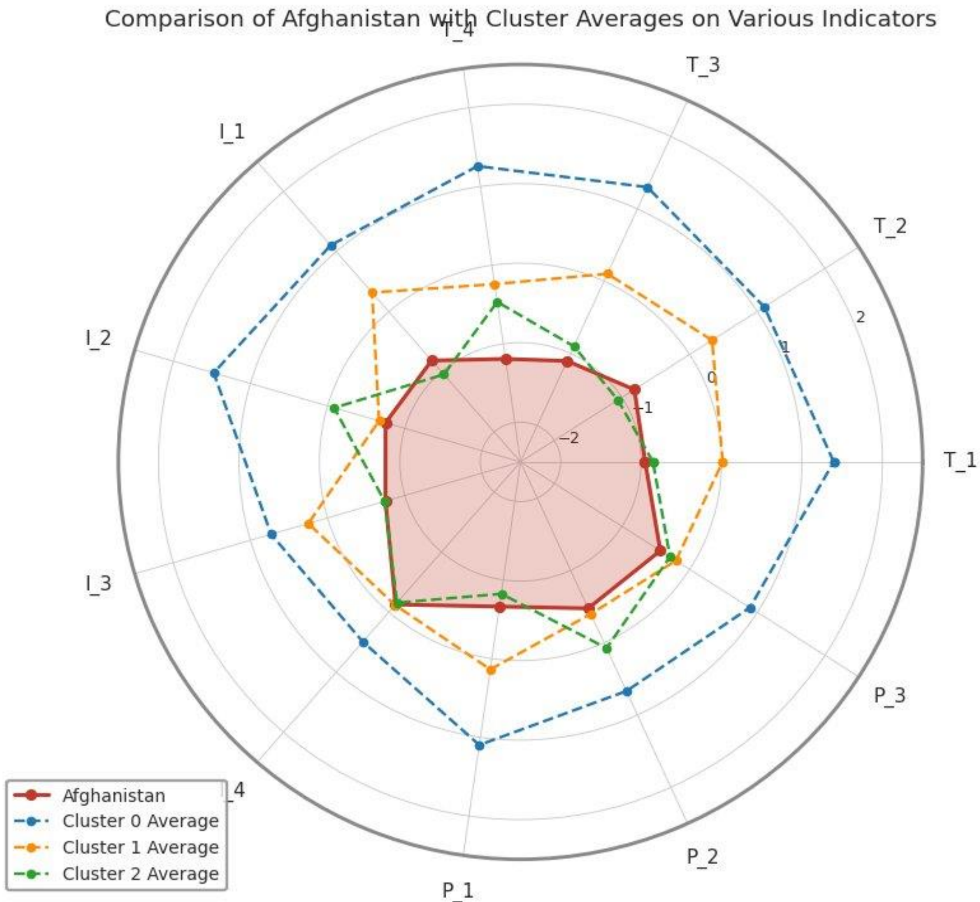


Figure 11 Indicator profile of Afghanistan vs all three cluster averages.

Afghanistan's profile (filled red region) sits well inside the Cluster 0 boundary across every indicator, and remains inside or close to the Cluster 2 boundary on most dimensions confirming the cluster assignment as consistent with the indicator profile rather than being driven by a small number of indicators. The profile shape highlights the relative compactness of underperformance: the country is broadly weak across the indicator set rather than catastrophically weak on a few dimensions, which has implications for intervention design favouring broad-spectrum capacity-building over narrowly targeted single-sector interventions.

6. Member States Covered

The full framework applies the methodology described in Sections 2 and 3 to twenty-eight member states with tables and visualisations structured in a different way than this report. The list is preserved below; per-country snapshots beyond the Afghanistan illustration in Section 5 are redacted from this version.

#	Member State	#	Member State	#	Member State
1	Afghanistan	11	Ethiopia	21	Palestine
2	Botswana	12	Gambia	22	Somalia
3	Burundi	13	Guyana	23	Togo
4	Cambodia	14	Jordan	24	Tunisia
5	Comoros	15	Lesotho	25	Türkiye
6	Congo	16	Mozambique	26	Uganda
7	Costa Rica	17	Nicaragua	27	Yemen
8	Côte d'Ivoire	18	Niger	28	Zimbabwe
9	Djibouti	19	Nigeria		
10	Eswatini	20	Palau		

7. Summary

The framework presented here serves two complementary functions. As an impact measurement instrument, it establishes a comparable, reproducible baseline for technological readiness across heterogeneous member states, against which future progress can be tracked at the indicator, pillar, and composite level. As a decision-support model, it produces country-specific, cluster-relative Priority Matrices that translate composite scores into actionable intervention sequencing recommending where each country's most consequential and feasible gaps lie, anchored against peers facing comparable structural conditions rather than against a uniform global mean.

The methodological pipeline Z-score standardisation, MICE iterative imputation, GMM clustering, ANOVA validation, PCA-derived weights, and rank-based priority assignment is conventional, transparent, and reproducible. The substantive contribution is the integration of these components into a single, internally consistent decision-support framework tailored to the data constraints and policy needs of Global South member states, together with the design choice of cluster-relative rather than global-mean benchmarking, which makes the Priority Matrix outputs operationally useful rather than uniformly damning.